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# **Did Remote Work Narrow the Motherhood Penalty?**

**Evidence on the Intensive Margin from Israeli Labor Force  
Survey Microdata, 2017–2023**

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## Abstract

The post-pandemic shift to remote work is a persistent shock to the flexibility of employment, the job attribute mothers have historically paid most for. We ask whether the rise of work from home (WFH) narrowed the motherhood penalty in usual weekly hours among employed women in Israel. We use Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023 to estimate difference-in-differences (DiD) and triple-differences (DDD) models of usual weekly hours, with an Israeli-calibrated occupational WFH-exposure index as the third difference. Employed mothers’ hours did not change on average (0.23 hours), but they rose where the work can be done from home: the triple interaction is 3.22 hours per unit of exposure, or 1.71 hours a week between the mean exposures of the most and least teleworkable quartiles of occupations. Because exposure varies across only forty occupations, we report analytic, permutation and wild cluster bootstrap inference, and the gradient survives all three at the 10% level or better. It is largest for mothers of children under five, and fathers show the opposite gradient. The employment margin is estimated too imprecisely to support a comparable conclusion. The evidence indicates that remote work raises intensive-margin labor supply for mothers in teleworkable occupations specifically, rather than reducing the motherhood penalty across the board.

## 1 Introduction

The shift to remote work is a rare natural experiment in workplace flexibility. Unlike the acute disruption of 2020, work from home has proven to be a persistent feature of the post-pandemic labor market (Barrero, Bloom and Davis, 2021), and flexibility has been identified as the last major driver of the remaining gender pay gap (Goldin, 2014). Temporal and locational flexibility is a good mothers pay for, in lower wages or slower promotion, so a shock that lowers the cost of supplying it should be detectable in mothers’ labor supply, and for mothers already employed most directly in how many hours they supply. This paper asks whether the post-COVID shift to remote work altered the motherhood penalty in *usual weekly work hours* among employed women in Israel, using Central Bureau of Statistics (CBS) Labor Force Survey microdata for 2017–2023.

The question has been asked once before, for the United States, where work from home reduced the motherhood penalty on the extensive margin and on income (Harrington and Kahn, 2025): a 10% rise in WFH raises mothers’ employment by roughly 0.78 percentage points relative to other women, and employed mothers’ incomes rise by about 1.3%. We ask it of a different country and a different margin: whereas the US evidence concerns whether mothers hold jobs and what they earn, we study how many hours the mothers who hold jobs supply, and in doing so provide, to our knowledge, the first estimate of the Israeli motherhood penalty in hours conditional on working. Israel is a natural setting for

the intensive-margin question. It combines a productive, internationally integrated labor market with the OECD’s highest total fertility rate, 2.87 children per woman in 2024 against an OECD average of 1.40 (OECD, 2025), so an unusually large share of its female workforce is at any time the mother of a young child, and an intensive-margin response to flexibility has first-order consequences for aggregate labor input. Between 13% and 16% of employed Israeli women aged 25–59 usually worked from home in 2021–2023, the first years in which the survey asked. We also estimate the employment margin, but for reasons of statistical power documented in Section 6.6 we treat it as secondary.

Detecting an hours response to flexibility requires a shift in its cost that is large, sudden and unrelated to any one mother’s circumstances, and the pandemic supplied exactly that. Our design therefore exploits the pandemic as the moment at which remote work became a mainstream practice. We compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 (Section 3). A difference-in-differences on this contrast estimates the change in the hours penalty. A triple-differences model that interacts it with an occupation’s WFH exposure tests whether remote work is the *mechanism*, by asking whether the change is larger in jobs that can actually be done from home. Its identifying assumption concerns how the motherhood gap *within* occupations would have evolved across exposure levels absent the shift; pre-period event studies probe that assumption rather than verify it.

Three findings emerge. The first is a null: averaged over all employed women, the motherhood penalty in hours is no smaller after the pandemic than before it. The second is that the average conceals a gradient. In occupations whose work can be done from home, the hours gap between mothers and childless women closed by an amount of the same order as the penalty itself, while in occupations tied to a workplace it did not close at all and, if anything, widened slightly; the response is concentrated in the most teleworkable quartile of occupations. The third is that the gradient behaves as a flexibility mechanism should. It is largest for mothers of children under five, for whom the child-care time constraint binds most tightly, and it declines as the youngest child ages; fathers in the same occupations show a gradient of the opposite sign, which rules out a shock common to everyone in teleworkable jobs and is the pattern a within-household reallocation of paid and care time would produce, a reading the data permit rather than one the design identifies. Because the exposure score varies across only forty occupations, we report analytic, permutation and wild cluster bootstrap inference side by side, and the strength of the statistical conclusion depends on which is used (Section 6.2). The estimate survives specification, selection, outcome-coding and leave-one-occupation-out checks and a test of sorting across occupations. It does depend on the calibration of the exposure index in one identifiable way: reclassifying teaching, which US task content scores as teleworkable but Israeli schools kept in person, is what separates it from the weaker, insignificant estimate the one fully pre-treatment exposure measure yields (Section 6.3).

Relative to the US evidence, the paper moves the question to the intensive margin, where the penalty operates through how much mothers who hold jobs can work rather than through whether they hold them; it measures exposure with an index calibrated to realized Israeli remote work rather than imported unchanged; and it treats the small number of occupation clusters as part of the evidence rather than as a nuisance. Section 2 reviews the literature, Sections 3 and 4 describe the data and the raw patterns, Section 5 sets out the design, Section 6 reports the results, and Sections 7–9 interpret them and state their limits.

## 2 Literature Review

Three questions organize the literature this paper draws on. Why does a motherhood penalty persist in labor markets where women’s education and experience match men’s, and is the inflexibility of jobs the reason? Does remote work, the largest shift in that inflexibility in decades, raise or lower the hours mothers supply, and does it narrow or widen gender gaps? And how can an economy-wide change in remote work be measured, and its effect identified, when no one is assigned to it at random? We take the three in turn, then collect what is known for Israel and state what this paper adds.

**The motherhood penalty and the flexibility mechanism.** The child penalty is one of the most robust facts in labor economics: across countries with very different family policies, women’s earnings fall sharply and persistently relative to men’s after a first birth, through reduced hours, exit from employment and moves into family-friendly, lower-paying jobs, and the size of the penalty tracks gender norms rather than policy (Kleven, Landais and Sogaard, 2019; Kleven et al., 2019). Hours are one of the channels, which is why an intervention that changes the time cost of combining work and care could move the intensive margin even where it does not move employment. The remaining gap has been located in the structure of jobs (Goldin, 2014): firms in many occupations pay disproportionately for long, contiguous, on-site hours, and workers who cannot supply them, disproportionately mothers, pay a penalty out of proportion to the hours they forgo. Remote work lowers the cost of supplying a given number of hours, by removing the commute, letting hours be split around care and permitting a virtual presence in demanding professions. Both premises have empirical support: workers, and women in particular, give up wages for the option to work from home (Mas and Pallais, 2017), and the skilled gender gap is concentrated where long hours are rewarded and narrows where substitutes for household time are cheap (Cortés and Pan, 2019). Remote work may also cut the other way, through lower visibility and slower advancement, so the sign is an empirical question. Goldin’s framework says where to look for it: in the gradient of the response across occupations by teleworkability, not in its average.

**What remote work does to hours and to gender gaps.** Remote work is now a settled feature of rich-country labor markets rather than a pandemic expedient, accounting for roughly a quarter of paid workdays in the United States since 2021 (Barrero, Bloom and Davis, 2021, 2023), and the time it frees is split between paid work, care and leisure rather than returned to the employer in full (Aksoy et al., 2023). Whether it raises hours is therefore not settled by the commute alone, and the direct evidence points both ways. Under randomized home working and in the pandemic move home, time worked rose, in the latter case without output keeping pace (Bloom et al., 2015; Gibbs, Mengel and Siemroth, 2023); against this, time-use data show that on remote days part of the saved time goes to family and, for women, to household production, and that teleworking mothers face more interruptions to the workday (Pabilonia and Vernon, 2022). Those findings bracket the question this paper asks of mothers: whether the time remote work frees is returned to paid hours or absorbed by care. Two further regularities shape how the question can be asked. First, who selects into remote work differs sharply from what remote work does to a given worker (Emanuel and Harrington, 2024), so a credible design must measure *exposure* to remote work from the content of jobs rather than from realized remote-work status. Second, the effect on gender gaps depends on when it is measured. Under randomized or voluntary flexibility, household tasks shift toward men (Angelici and Profeta, 2024); under the acute 2020 shock, with schools closed, women in WFH-feasible jobs absorbed the larger share of the childcare that resulted, as was anticipated at the time (Alon et al., 2020; Adams-Prassl et al., 2020). Evidence on the post-acute period, once schools reopened and remote work became a job attribute rather than a lockdown condition, is much thinner, and it is that period this paper studies. The one study of remote work and the motherhood penalty in it (Harrington and Kahn, 2025) finds that WFH raised US mothers’ employment and income, with the employment effect concentrated in fields with high returns to hours and inflexible time demands. Hours are one channel of its income result, which makes our estimate its closest analogue while leaving open whether an intensive-margin effect exists on its own, and in a different country.

**Measurement and identification.** Because remote work is chosen, the literature measures it in two ways: *ex ante*, by whether an occupation’s tasks can be performed entirely at home, a criterion over a third of US employment meets (Dingel and Neiman, 2020); or *ex post*, by the share of workers in an occupation who actually work from home. The first is immune to selection but fixed by task content, so it scores an occupation identically in every country; the second reflects local practice but is itself a labor-market outcome. The two diverge wherever institutions overrode task content, as they did for teaching, which task content scores as fully teleworkable but which Israeli schools kept in person throughout the survey years. Our exposure index starts from the task-based classifica-

tion mapped to two-digit ISCO-08 codes and calibrates it against realized Israeli remote work (Section 3.2), for which a pre-pandemic descriptive baseline exists (Madhala and Bental, 2021). On identification, a continuous exposure score places the design in the continuous-treatment difference-in-differences family, whose recent results caution that a single slope compares outcomes across exposure levels, requires parallel trends across those levels rather than within them, and averages the comparisons with weights that need not be positive (Callaway, Goodman-Bacon and Sant’Anna, 2024). Section 5 takes up the two results that bear on reading our coefficient.

**What is known for Israel.** The Israeli evidence concerns other margins and other shocks. The gender wage gap is driven mainly by the sorting of women into lower-paying firms, even within industry, with pay differences within firms contributing little (Buzaglo-Baris, 2023), a reminder that any occupation-level exposure measure captures only part of the relevant heterogeneity. Motherhood penalties in employment and earnings differ across ethno-religious groups (Budig, Kraus and Levanon, 2023), and the 2020 lockdown reallocated paid and unpaid work within households in gender-unequal ways (Yaish, Mandel and Kristal, 2021). There is, to our knowledge, no published Israeli estimate of the motherhood penalty in hours conditional on working, so the level we report has no domestic benchmark, and no Israeli study asks whether remote work changed any margin of the penalty.

**What this paper adds.** Against these strands the contribution is specific. To the flexibility literature it brings the first estimate of the Israeli motherhood penalty in hours conditional on working, and a direct test of whether remote work changed that penalty on the intensive margin, a margin the US evidence reaches only indirectly through income. To the remote-work literature it brings post-acute rather than lockdown-period evidence, from an economy where mothers of young children are an unusually large share of the female workforce. To the measurement question it shows, through the teaching occupations, that calibrating a task-based exposure index to local practice can decide the result (Section 6.3), and it treats the small number of occupation clusters as part of the evidence, reporting three inference procedures side by side rather than one.

## 3 Data and Institutional Context

### 3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017–2019 and 2021–2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is a *mother* if she has

any child under 17 in the household and childless otherwise; *Post* equals one for survey years 2021 and later. The comparison group therefore includes women whose children are grown and women who become mothers within the window (Section 8). The year 2020 is excluded because its arrangements were emergency improvisation rather than the settled remote-work practice whose effect we want to measure, and because no 2020 CBS extract exists for this project.

The primary outcome is *usual weekly work hours*, recorded by the CBS in bins and assigned each bin’s median (Appendix A gives the coding and a 2017 recording correction). Hours are the only intensive-margin outcome the survey supports: the extract records how a wage is paid but not its amount. Hours are defined for women who are employed *and* worked in the survey’s reference week, in every year, a harmonization made necessary by a different zero-hours recording convention in the 2017 extract (Appendix A); it makes the hours population identical across years at the cost of about 10% of each year’s employed sample and narrows the estimand to usual hours among women who were working (Section 8). The secondary outcome is a binary employment indicator (unemployed and not-in-labor-force are both zero). All regressions include the same categorical controls: marital status, religion, age group, district of residence and highest educational attainment (six groups); age enters as the CBS age-group code because the extract contains no continuous age. CBS survey weights are deliberately not applied in any outcome regression (Section 8). The replication code is available in the project’s Git repository.<sup>1</sup>

### 3.2 Constructing WFH exposure

Realized WFH status is itself a labor-market outcome, jointly determined with hours, so the mechanism test needs an occupation-level measure of *exposure* to remote work that the individual does not choose. We build three, each reported.

*External.* The Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) at the two-digit ISCO-08 level. This is pre-period by construction, but it is a US task-based measure and misclassifies occupations where Israeli practice diverged: teaching scores 0.966 on it, yet Israeli schools were fully in person throughout the 2022–23 survey years (Appendix B).

*Calibrated (primary).* The rule that selects the primary measure is fixed ex ante: the score the headline uses must be anchored on pre-period task content, so that it cannot be built from the post-2021 hours it is meant to explain, and must not place in the treated tail occupations that Israeli institutions kept in person. The external index fails the second condition and the realized index the first; the calibrated index is the only one of the three that meets both, and the other two are reported as bounds in Section 6.3. It replaces the external score with the realized 2022–23 Israeli WFH share only where the

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<sup>1</sup><https://github.com/NitzanZacharia/labor-econ-remote-work-analysis-hours.git>

two are far apart and the gap is statistically distinguishable from noise; ten of the forty occupations are swapped, teaching and clerical support among them (rule, scores and fit in Appendix B). Because the swap test draws on 2022–23 data, inside the post-period, the calibrated measure is *partly post-treatment*, a documented compromise rather than a fully pre-treatment index; Section 6.3 tests whether the calibration is what produces the result.

*Realized.* The realized Israeli WFH share by occupation, anchored to 2021. Post-treatment by construction, it is reported as a bound.

A fourth measure serves three roles. For the employment DDD, occupation is undefined for the non-employed, so that specification uses a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religion and continent of birth; 8,884 cells), equal to the mean of the calibrated occupation score over the cell’s pre-period occupational composition and defined for employed and non-employed women alike. The same index stratifies the selection correction of Section 5.4, and because a woman’s post-2021 occupation cannot move it, it also bounds occupational sorting in Section 6.3. Its dispersion is a third of the occupation-level index’s, so coefficients on the two are compared per standard deviation, never per unit.

## 4 Descriptive Statistics

### 4.1 The sample in summary

Table 1 describes the sample on either side of the treatment/control split. Mothers work *fewer* hours than childless women when they work (38.68 versus 40.06) while being slightly *more* likely to be employed (0.78 versus 0.76), so the level of the penalty is visible on the intensive margin; whether it changed after 2021 is not something the table can show. The two groups are imbalanced in ways the controls must absorb: mothers are the better-educated group (48% with an academic degree against 40%), overwhelmingly married (85% against 47%), and about 0.44 of an age-group code younger, an imbalance that tracks WFH exposure (Appendix Table A4).

### 4.2 The raw gap and its exposure gradient

Mothers work 38.54 hours per week before 2021 and 38.84 after, a gain of 0.31 hours; childless women move from 40.04 to 40.08. The difference between those two changes, 0.27 hours, is the difference-in-differences of equation (1) without a single control, and the regression DiD of Table 2, 0.228 (SE 0.181), is within four hundredths of an hour of it. Mothers remain well below childless women in both periods, and the raw gap widened by 0.3 hours over the three pre-period years, within sampling error (Appendix Figure A1).

Table 1: Summary statistics: women aged 25–59, by mother status

|                                              | Childless women | Mothers | Difference (SE) |
|----------------------------------------------|-----------------|---------|-----------------|
| <i>Panel A: outcomes</i>                     |                 |         |                 |
| Observations                                 | 133,646         | 239,095 |                 |
| Employment rate                              | 0.7574          | 0.7828  | 0.0254 (0.0031) |
| Usual weekly hours (ref.-week workers), mean | 40.06           | 38.68   | −1.38 (0.09)    |
| standard deviation                           | [12.12]         | [11.62] |                 |
| observations with usual hours                | 93,982          | 164,193 |                 |
| <i>Panel B: characteristics</i>              |                 |         |                 |
| Age-group code (CBS 3–7), mean               | 5.40            | 4.96    | −0.44 (0.01)    |
| Arab (%)                                     | 17.73           | 17.78   | 0.05 (0.33)     |
| <i>Marital status (%)</i>                    |                 |         |                 |
| Married                                      | 46.55           | 84.57   | 38.02 (0.35)    |
| Married, living separately                   | 3.58            | 0.84    | −2.74 (0.11)    |
| Divorced                                     | 13.13           | 7.70    | −5.42 (0.24)    |
| Widowed                                      | 2.75            | 0.99    | −1.76 (0.11)    |
| Single, never married                        | 34.00           | 5.90    | −28.10 (0.31)   |
| <i>Religion of household head (%)</i>        |                 |         |                 |
| Jewish                                       | 73.08           | 79.50   | 6.42 (0.36)     |
| Christian                                    | 7.12            | 2.23    | −4.89 (0.18)    |
| Muslim                                       | 13.78           | 14.89   | 1.11 (0.30)     |
| Druze                                        | 1.88            | 1.64    | −0.24 (0.12)    |
| Other                                        | 4.15            | 1.75    | −2.40 (0.14)    |
| <i>District of residence (%)</i>             |                 |         |                 |
| Jerusalem                                    | 8.76            | 12.31   | 3.56 (0.26)     |
| North                                        | 17.52           | 16.58   | −0.94 (0.32)    |
| Haifa                                        | 14.38           | 11.92   | −2.45 (0.28)    |
| Center                                       | 23.99           | 26.52   | 2.53 (0.36)     |
| Tel Aviv                                     | 21.55           | 14.90   | −6.65 (0.33)    |
| South                                        | 11.66           | 12.37   | 0.72 (0.27)     |
| Judea and Samaria                            | 2.15            | 5.39    | 3.24 (0.15)     |
| <i>Education (%)</i>                         |                 |         |                 |
| Below high school                            | 11.37           | 6.63    | −4.74 (0.23)    |
| High school, no matriculation                | 12.70           | 11.95   | −0.74 (0.24)    |
| Matriculation (Bagrut)                       | 22.73           | 18.82   | −3.90 (0.29)    |
| Post-secondary, non-academic                 | 11.50           | 13.68   | 2.18 (0.24)     |
| Academic degree (BA/MA/PhD)                  | 40.24           | 47.97   | 7.73 (0.39)     |
| Other / no certificate                       | 1.47            | 0.95    | −0.52 (0.07)    |

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59; a mother has any child under 17 in the household. Weekly hours are conditional on working in the reference week, with the standard deviation in brackets and their own observation count. The difference column is mothers minus childless women, with a standard error clustered by individual. All figures are *unweighted*. The Arab share is the CBS population-group code used for the subgroup analysis; religion is a separate item recorded for the household head. Marital status, religion, district, age group and education are the five controls in every regression.

Figure 1 computes the same difference-in-differences separately within each quartile of occupational WFH exposure. The bottom three quartiles return  $-0.32$ ,  $0.27$  and  $-0.02$  hours, none distinguishable from zero; the top quartile returns  $1.55$  hours (95% CI  $[0.74, 2.36]$ ). The near-zero average is therefore a weighted average of three nulls and one substantial positive difference. The figure is a description rather than an independent causal test: occupational composition and time-varying confounders can differ across quartiles, which is what the regressions of Section 6 address. The response is not linear in exposure: the bottom three quartiles span exposure values of  $0.00$  to  $0.30$  and the top quartile  $0.30$  to  $0.75$ , so most of the regressor’s dispersion, and the identifying contrast, sits in one tail.



Figure 1: Raw hours difference-in-differences by quartile of WFH exposure

*Notes:* Women aged 25–59 who worked in the reference week and carry a matched occupation code. Each point is the raw difference-in-differences within one quartile of the occupation-level calibrated exposure measure, from unweighted cell means with no controls; bars are 95% confidence intervals clustered by individual. Breakpoints are computed on pre-period rows and applied to all years; each quartile’s exposure range is printed beneath its tick. The 1.55 hours in the top quartile is that quartile’s own difference-in-differences; the raw contrast between the top and bottom quartiles is  $1.55 - (-0.32) = 1.87$  hours.

## 5 Empirical Strategy

### 5.1 Difference-in-differences

For outcome  $Y_{it}$  (weekly hours conditional on working that week, or employment) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where  $X_{it}$  is the control vector of Section 3 and  $\beta_3$  is the differential change in the outcome for mothers relative to childless women after the shift to remote work.

## 5.2 Triple differences

The flexibility framework (Goldin, 2014) yields one testable hypothesis: after 2021, mothers' usual hours conditional on working rise relative to childless women's in proportion to their occupation's WFH exposure, with no response where the work cannot be done from home. The exposure gradient, not the average change, is the object of interest, so we interact the DiD with occupational WFH exposure:

$$\begin{aligned} Y_{it} = & \alpha + \delta_1 \text{Mother}_i + \delta_2 \text{Post}_t + \delta_3 \text{WFH\_Exposure}_{j(i)} + \delta_4 (\text{Mother}_i \times \text{Post}_t) \\ & + \delta_5 (\text{Mother}_i \times \text{WFH\_Exposure}_{j(i)}) + \delta_6 (\text{Post}_t \times \text{WFH\_Exposure}_{j(i)}) \\ & + \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH\_Exposure}_{j(i)}) + X'_{it} \gamma + \varepsilon_{it}, \end{aligned} \quad (2)$$

where  $j(i)$  is individual  $i$ 's occupation.  $\delta_7$  asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupations;  $\delta_6$  absorbs any post-2021 change in teleworkable occupations common to mothers and childless women,  $\delta_5$  the pre-existing level of the penalty across occupations, and  $\delta_4$  is the post-shift change for mothers at *zero* exposure, not an average effect. The DDD requires a single parallel-trends assumption (Olden and Møen, 2022): that the mother/non-mother gap in high-versus low-exposure jobs would have evolved in parallel absent the shift to remote work. Insignificant pre-period coefficients are evidence about that assumption, not a verification of it. For the hours DDD,  $\text{WFH\_Exposure}_j$  is the calibrated occupation-level score and the sample is employed women matched to a two-digit occupation; for the employment DDD,  $\text{WFH\_Exposure}$  is the cell-based index, the sample is all women, and a  $\text{Mother} \times \text{age-group}$  interaction is added for the pre-period age imbalance.

Equation (2) makes each lower-order occupation-level difference linear in exposure, an unnecessary restriction with forty occupations, and comparing effects *across* exposure levels, which is what a single slope does, requires the stronger condition that occupations at different exposure levels would have followed the same path under any common level of exposure, and the linear slope is a weighted average of such comparisons whose weights need not be positive (Callaway, Goodman-Bacon and Sant'Anna, 2024). We therefore estimate two relaxations alongside it. The *saturated* specification replaces every lower-order term with fixed effects,

$$Y_{it} = \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH\_Exposure}_{j(i)}) + X'_{it} \gamma + \lambda_{j(i),t} + \mu_{j(i),\text{Mother}_i} + \nu_{\text{Mother}_i,t} + \varepsilon_{it}, \quad (3)$$

where  $\lambda_{jt}$ ,  $\mu_{j,m}$  and  $\nu_{m,t}$  are occupation-by-year, occupation-by-mother and mother-by-year effects, which absorb every lower-order term of equation (2) and any occupation-

specific year shock. The *binned* specification replaces  $\text{WFH\_Exposure}_{j(i)}$  with indicators for the pre-period exposure quartile the occupation falls in, the bins of Figure 1, so that the top-versus-bottom contrast carries a standard error. For each DDD we also report the minimum detectable effect on the reference distribution the model’s inference uses,  $\text{MDE} = \text{SE}(t_{0.975} + t_{0.80})$ , which is  $2.87 \times \text{SE}$  with 39 degrees of freedom and  $2.80 \times \text{SE}$  under the normal, so that a null can be read as “genuinely null” or “underpowered.”

### 5.3 Inference

The survey is a rotating panel in which most respondents appear about four times, so equation (1) and every descriptive interval are clustered by individual (Appendix A gives the panel counts). For the hours DDD the regressor is constant within a two-digit occupation, and standard errors are clustered at that level, leaving forty clusters; the reported  $p$ -values use a  $t$  reference distribution with  $G - 1 = 39$  degrees of freedom and a cluster-count adjustment to the variance. Forty is few enough that we do not rest on that approximation: for the headline coefficient and for the pre-period coefficients of the triple-difference event study we also report wild cluster bootstrap  $p$ -values, with Rademacher weights, the null imposed and 9,999 draws (Roodman et al., 2019; Fischer and Roodman, 2021), and Section 5.5 adds a permutation test that relies on no asymptotic approximation at all. The same bootstrap is applied to every occupation-clustered robustness and subgroup refit and printed beside its analytic  $p$ -value, since a check starred on a test that over-rejects would be weaker evidence than it looks. The three procedures answer related but different questions, and where they disagree the disagreement is part of the evidence. The joint pre-period test remains the analytic Wald  $F$ , since a joint bootstrap test is not available in the implementation used. The employment DDD is clustered on the age-by-education-by-district cell (about 210 clusters).

### 5.4 Selection into employment: Lee bounds

Hours are observed only for women who are working, and working is itself an outcome of the same treatment. We therefore report trimming bounds (Lee, 2009): the share of post-period mothers observed with hours in excess of a parallel-trends counterfactual is trimmed from the top of that cell’s hours distribution for a lower bound and from the bottom for an upper bound, the model is refit, and the Imbens–Manski interval for the identified set (Imbens and Manski, 2004) is reported. For the DDD the trimming is done within quartiles of the cell-based index, which is defined for the non-employed. Appendix C gives the construction and its assumptions. The bounds address excess selection under monotonicity; they cannot address selection that violates it, nor differences in absence unrelated to excess selection.

## 5.5 Diagnostics, falsification, and the fathers’ comparison

We estimate event-study versions of equation (1) with  $\text{Mother} \times \text{year}$  interactions (reference year 2019) for both outcomes and test the pre-period coefficients jointly; because those test the difference-in-differences assumption rather than the triple-differences one, we also estimate the event-study version of equation (2) itself, with  $\text{Post}$  replaced by survey-year interactions at every level, and apply the same joint test to its pre-period coefficients.

The falsification test is a permutation test on the exposure score. Under the sharp null that teleworkability is unrelated to the post-2021 change in the motherhood gap, the forty occupation scores are exchangeable, so we reassign them at random across the forty occupation codes 999 times, holding every woman’s occupation and the mother/period structure fixed, refit equation (2), and record the cluster-robust  $t$ -statistic of the triple interaction; the permutation  $p$ -value is the share of draws, counting the observed one, whose  $|t|$  reaches the observed value (Phipson and Smyth, 2010). Because occupation is observed when the outcome is, we also check whether remote work drew mothers into teleworkable occupations after 2021, with a difference-in-differences that takes the exposure score itself as the outcome, and bound the channel by re-estimating the hours DDD on the pre-period cell index of Section 3.2, which a woman’s later occupation cannot move. The remaining checks, collected in Table 3, re-estimate the hours DDD with the exposure measure, the age control, the sample, the outcome coding and the inference procedure varied in turn, and re-estimate it by the age of the youngest child, separately for Jewish and Arab women, and on the male subsample with “mother” read as “father.” The last is a comparison, not a placebo: fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a negative gradient would be evidence on household reallocation rather than on a confound. Coefficients across disjoint subsamples are compared with two-sample  $z$ -tests.

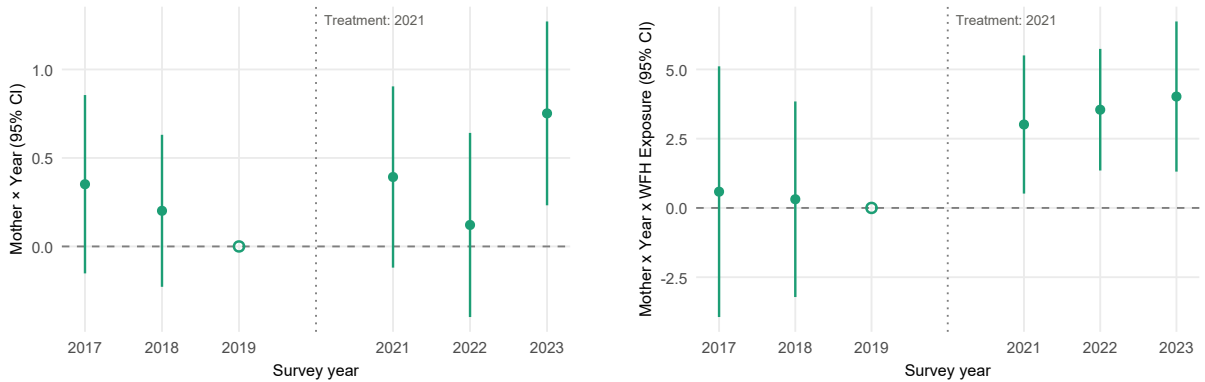
## 6 Results

Three findings organize this section. The average motherhood penalty in weekly hours did not change after the shift to remote work. It did narrow where the work can be done from home: the post-2021 change in the mother/childless gap rises with an occupation’s WFH exposure, with the response concentrated in the most teleworkable quartile of occupations. That gradient survives the design’s checks on selection, sorting, calibration and specification, is largest for mothers of young children and reversed in sign for fathers, while the employment margin is too imprecisely estimated to support a comparable conclusion. Tables 2–5 and Figure 2 carry the estimates; the text reports the pattern. Significance is denoted \*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , \*  $p < 0.05$ , ·  $p < 0.1$ , with standard errors in parenthe-

ses. Stars follow the analytic occupation-clustered  $t$ -test; because that test over-rejects with forty clusters (Section 5.3), Tables 3–5 also print a wild cluster bootstrap  $p$ -value beside every occupation-clustered triple interaction.

## 6.1 No average change in the hours penalty

Column (1) of Table 2 estimates equation (1). Mothers work 1.739 fewer hours per week than childless women, and that penalty did not measurably change after 2021: Mother  $\times$  Post is 0.228 (SE 0.181); its 95% confidence interval,  $[-0.13, 0.58]$ , excludes average gains above about half an hour a week, and the design could have detected an effect of about 0.51 hours. Panel (a) of Figure 2 supports the comparison: the pre-period coefficients are indistinguishable from zero and pass a joint Wald test ( $F(2, 63, 198) = 0.949$ ,  $p = 0.387$ ), and the post-period path is flat apart from a larger 2023 coefficient. Survey-year effects in place of the single Post dummy, and the selection bounds of Appendix C, leave the reading unchanged. Nothing that follows rests on this average.



(a) DiD: Mother  $\times$  year

(b) DDD: Mother  $\times$  year  $\times$  WFH\_Exposure

Figure 2: Event studies for weekly hours, reference year 2019

*Notes:* Vertical bars are 95% confidence intervals; the hollow point at 2019 is the omitted reference year, and the gap between 2019 and 2021 is the excluded 2020 survey year. Panel (a): women working in the reference week ( $N = 251,857$ ), standard controls, standard errors clustered by individual. Coefficients (SE): 2017, 0.352 (0.257); 2018, 0.201 (0.219); 2021, 0.392 (0.261); 2022, 0.121 (0.265); 2023, 0.752\*\* (0.265). Joint test on 2017–2018:  $F(2, 63, 198) = 0.949$ ,  $p = 0.387$ . Panel (b): sample, controls, exposure and clustering as in column (2) of Table 2 ( $N = 246,326$ , 40 clusters), with the year main effects and both two-way year interactions alongside the plotted triple interaction. Coefficients (SE), hours per unit of exposure: 2017, 0.587 (2.237); 2018, 0.313 (1.745); 2021, 3.011\* (1.233); 2022, 3.545\*\* (1.085); 2023, 4.020\*\* (1.340). Joint test on 2017–2018:  $F(2, 39) = 0.041$ ,  $p = 0.960$ ; wild cluster bootstrap  $p$ -values (Rademacher, 9,999 draws) 0.838 for 2017, 0.884 for 2018 and 0.852 for their sum.

## 6.2 Hours rose where the work can be done from home

Column (2) of Table 2 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.224\*\* (SE 1.022): among working mothers, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations. That is 0.64 hours per standard deviation of the index (0.199), and 1.71 hours between the mean exposures of the bottom and top quartiles of Figure 1, the pattern that figure shows in the raw cell means. The implied change at zero exposure is  $-0.539$  (SE 0.336), so mothers in jobs that cannot be done from home gained nothing, and the gradient is net of whatever happened to hours in teleworkable occupations for all women. What changed after 2021 is not mothers' hours in general but mothers' hours where remote work was possible.

Columns (3) and (4) relax the functional form and tell the same story. With the three sets of fixed effects of equation (3) absorbing every lower-order term, the triple interaction is 2.876 (SE 0.950), a tenth below the pooled estimate and still significant at the 1% level. With quartile indicators in place of the score, the post-2021 change in the gap relative to the least teleworkable quartile is positive but insignificant in the second and third quartiles and 1.844 hours (SE 0.595,  $p < 0.01$ ) in the top one: a step at the top of the exposure distribution more than a smooth gradient. That quartile holds six of the forty occupations, but the pooled and saturated columns, which use all forty, agree with it, and no single occupation drives it (Section 6.4).

Panel (b) of Figure 2 is the event-study version of equation (2), the pre-trend test the triple difference itself requires (Section 5.5). The pre-period coefficients are near zero and jointly insignificant ( $F(2, 39) = 0.041$ ,  $p = 0.960$ ; wild cluster bootstrap  $p = 0.85$  on their sum), and the post-period path steps up in 2021 and rises through 2023, every year straddling the pooled estimate. The test has limited power: with forty clusters the pre-period standard errors are of the same order as the effect, so each pre-period interval contains values as large as the estimate itself. It shows no sign of a violation; it does not rule one out.

How precisely the gradient is estimated depends on the inference procedure, and we report three. The analytic standard error, clustered by occupation on a  $t_{39}$  reference, puts the estimate 3.15 standard errors from zero ( $p = 0.003$ ). The two procedures built for forty clusters are less emphatic. The permutation test gives  $p = 0.032$ : 31 of 999 random reassignments of the forty scores across occupations produce a  $|t|$  at least as large as the observed one (Appendix Figure A2), so a design with this cluster structure produces a gradient this large about three times in a hundred when exposure carries no information. The wild cluster bootstrap gives  $p = 0.056$ , with a 95% confidence set of  $[-0.26, 5.84]$ . We report all three rather than choosing among them.

Table 2: Intensive-margin DiD and DDD: weekly hours, reference-week workers

|                                            | (1) DiD              | (2) DDD              | (3) DDD, saturated | (4) DDD, quartiles   |
|--------------------------------------------|----------------------|----------------------|--------------------|----------------------|
| Mother $\times$ Post $\times$ WFH_Exposure |                      | 3.224**<br>(1.022)   | 2.876**<br>(0.950) |                      |
| Mother $\times$ Post $\times$ Q2 (vs. Q1)  |                      |                      |                    | 0.753<br>(0.483)     |
| Mother $\times$ Post $\times$ Q3 (vs. Q1)  |                      |                      |                    | 0.600<br>(0.485)     |
| Mother $\times$ Post $\times$ Q4 (vs. Q1)  |                      |                      |                    | 1.844**<br>(0.595)   |
| Mother $\times$ Post                       | 0.228<br>(0.181)     | -0.539<br>(0.336)    |                    | -0.472<br>(0.346)    |
| Mother $\times$ WFH_Exposure               |                      | -1.295<br>(1.053)    |                    |                      |
| Post $\times$ WFH_Exposure                 |                      | -0.548<br>(0.810)    |                    |                      |
| WFH_Exposure                               |                      | 1.128<br>(3.912)     |                    |                      |
| Mother                                     | -1.739***<br>(0.144) | -1.449***<br>(0.351) |                    | -1.527**<br>(0.452)  |
| Post                                       | -0.057<br>(0.148)    | 0.053<br>(0.273)     |                    | -0.130<br>(0.367)    |
| Constant                                   | 32.355***<br>(0.318) | 32.115***<br>(1.540) |                    | 31.581***<br>(2.051) |
| MDE for triple interaction ( $t_{39}$ )    |                      | 2.938                |                    |                      |
| SD of exposure, estimation sample          |                      | 0.199                |                    |                      |
| Triple interaction per SD of exposure      |                      | 0.640                |                    |                      |
| Occupations in Q1 / Q2 / Q3 / Q4           |                      |                      |                    | 16 / 7 / 11 / 6      |
| Fixed effects                              | None                 | None                 | Three sets         | None                 |
| Observations                               | 251,857              | 246,326              | 246,324            | 246,326              |
| $R^2$                                      | 0.040                | 0.041                | 0.135              | 0.048                |
| Clustering                                 | individual           | occupation (40)      | occupation (40)    | occupation (40)      |

Dependent variable: usual weekly work hours, women working in the reference week; standard controls in every column. Columns (2)–(4) use the calibrated occupation-level exposure index on column (1)’s sample less the rows with no matched two-digit occupation. Column (3) is equation (3), whose three sets of fixed effects (occupation  $\times$  year, occupation  $\times$  mother, mother  $\times$  year) absorb every lower-order term. Column (4) replaces the score with indicators for its pre-period quartile (Figure 1’s bins); its lower-order quartile terms are estimated but not shown, and the number of occupations per bin is listed because occupation is the cluster. Columns (2)–(4) use a  $t$  reference distribution with 39 degrees of freedom (Section 5.3), which the MDE also uses ( $\alpha = 0.05$ , power 0.80, multiplier 2.87); the wild-bootstrap and permutation  $p$ -values for column (2)’s triple interaction are in Table 3. Column (2)’s Mother  $\times$  Post is the effect at *zero* exposure, not an average effect. Coefficients and standard errors to three decimals throughout. No coefficient is dropped for collinearity in any column.

### 6.3 Selection, sorting, and the calibrated regressor

Three features of the design could produce an exposure gradient with no change in mothers' labor supply: selection into work, sorting of mothers across occupations, and an index calibrated on post-period data. Table 3 reports the tests.

**Selection into work.** Hours are observed only for women who work, and working is itself an outcome. Trimming the excess selection among post-period mothers within each quartile of the cell-based index (Appendix C) barely moves the estimate in either direction, and the Imbens–Manski 95% interval for the identified set, [1.021, 5.324], excludes zero.

**Sorting across occupations.** If remote work drew mothers into teleworkable occupations after 2021, the occupation score would itself respond, and it does not: as a dependent variable it gives a Mother  $\times$  Post under one hundredth of its standard deviation, the probability of being in the top quartile moves by 0.4 percentage points (SE 0.7), and the event-study version is flat in every year. The hours DDD on the pre-period demographic-cell index, which a woman's later occupation cannot move, is positive but imprecise (0.139 hours per standard deviation of that index, SE 0.159); that index is a far coarser measure of a woman's own teleworkability, so the sorting DiD is the informative test.

**Calibration.** The calibrated index draws on realized 2022–23 WFH shares, so the regressor could be absorbing post-period outcomes. The external Dingel–Neiman index (Dingel and Neiman, 2020), the only measure built entirely from pre-treatment information, gives 0.672 (SE 0.884), positive but insignificant and the weakest of the three exposure measures; the realized 2021-anchored index gives a larger, noisier estimate. Three checks hold the sample at all forty occupations. Calibrating on men only, so that no woman in the regression contributes to her own regressor, gives 3.869\*\*\* (SE 0.979). Keeping the external score but letting the ten occupations the calibration reclassifies carry their own Mother  $\times$  Post change restores a gradient in the external score at two thirds of the headline ( $p = 0.09$ ): the occupations the external index places in the treated tail did not respond. Dropping those ten, so that the two indices coincide on the thirty that remain, gives 4.309\*\* (SE 1.192). Reclassifying teaching alone, with the other thirty-nine occupations at their external scores, gives 1.934\*\*\* (SE 0.519), and moving the calibration's swap threshold from 0.30 to 0.90 leaves the estimate positive throughout, from 2.3 with only teaching and clerical support swapped to 3.9 (SE 2.8) with fifteen occupations swapped (Appendix Table A2). The result therefore requires reclassifying teaching, and that reclassification rests on an institutional fact rather than on any woman's hours: Israeli schools were fully in person in 2022–23 (Appendix B). The calibration cannot be made into a pre-treatment measure, and a reader who weights the external index above the others will read the headline as directionally supported rather than established.

## 6.4 Robustness of the gradient

The remaining checks in Table 3 vary the age and marital controls, the sample and the outcome coding, and two of them matter. Mothers and childless women differ in age, education and marital status in ways that themselves vary with exposure (Appendix Table A4), and such a gap is what the triple interaction is identified off. A Mother  $\times$  age-group interaction leaves the estimate essentially unchanged, but reweighting to balance pre-period age groups attenuates it by about 19% to 2.597\* (SE 1.050; bootstrap  $p = 0.05$ ), significant at the 5% level on the analytic test but below the design’s minimum detectable effect.

Marital status is the imbalance the age checks do not touch: 85% of mothers are married against 47% of childless women (Table 1), and the gap widens with exposure. Restricting the sample to married women gives 2.743 (SE 1.512,  $p = 0.08$ ), and adding marital status  $\times$  Post  $\times$  WFH\_Exposure to the specification gives 2.191 (SE 1.273,  $p = 0.09$ ): the gradient stays positive but loses conventional significance. Part of it, then, is plausibly the comparison group rather than mothers, if never-married childless women in teleworkable jobs, who make up a larger share of that group at high exposure, worked fewer hours after 2021. Neither row is a clean test. The interaction is partly over-control, since most mothers are married and the term is estimated partly off them, and the married subsample trades a marital confound for a life-stage one, since married childless women are older and often have grown children. We report both as the bounded version of a weakness the design cannot remove.

Everything else stays close to the headline. Excluding the 2023 survey year, whose final quarter includes the start of the October 2023 war (Section 8), gives 2.979\* (SE 1.170), while the plain hours DiD stays a null. Removing each of the forty occupations in turn (Appendix Figure A3) leaves the estimate between 2.415 and 3.580, within one standard error of the headline and significant at the 5% level on the analytic test in every refit ( $p \leq 0.017$ ). Across every specification in the table that keeps a calibrated index, the triple interaction lies between 2.2 and 4.3 hours. The wild cluster bootstrap column of Table 3 tempers the stars: the analytic test over-rejects with forty clusters, and on the bootstrap the calibrated-index rows range from  $p = 0.02$  (unswapped occupations, men-only calibration) to  $p = 0.13$  (marital interaction), with the remaining sample and outcome-coding rows all within two hundredths of the headline’s own 0.056.

Nor does the outcome coding drive it. Hours are a bin-median approximation and the effect is smaller than one bin width, but dropping the two imputed irregular-hours codes changes little, and because 35 and 40 hours are exact bin edges the same specification can be estimated as a linear probability model on either threshold: the probability of usually working at least 35 hours, and at least 40, rises by about 2 percentage points per standard deviation of exposure (1.9 and 2.1) for mothers relative to childless women after 2021.

The effect is a shift of mothers in teleworkable jobs across the full-time boundary.

Table 3: Hours DDD: sensitivity of Mother  $\times$  Post  $\times$  WFH\_Exposure

| Specification                                            | Coefficient    | SE    | $p$              | $p$ (boot) | $N$     | Clusters         |
|----------------------------------------------------------|----------------|-------|------------------|------------|---------|------------------|
| <i>Exposure measure</i>                                  |                |       |                  |            |         |                  |
| Calibrated (primary)                                     | 3.224**        | 1.022 | 0.0031           | 0.0558     | 246,326 | 40               |
| External (Dingel–Neiman)                                 | 0.672          | 0.884 | 0.4517           | 0.5780     | 246,326 | 40               |
| Realized (2021 anchor)                                   | 5.195*         | 2.363 | 0.0346           | 0.0278     | 246,037 | 36               |
| Calibrated on men only (12 of 40 swapped)                | 3.869***       | 0.979 | 0.0003           | 0.0244     | 246,326 | 40               |
| External index, teaching (ISCO 23) alone swapped         | 1.934***       | 0.519 | 0.0006           | 0.0020     | 246,326 | 40               |
| External index, swapped-occupation terms added           | 2.339          | 1.352 | 0.0915           | 0.2074     | 246,326 | 40               |
| Mother $\times$ Post $\times$ Swapped (same model)       | −1.347         | 0.926 | 0.1537           | 0.3041     | 246,326 | 40               |
| <i>Age and marital balance</i>                           |                |       |                  |            |         |                  |
| Age-interacted (+ Mother $\times$ age group)             | 3.205**        | 1.055 | 0.0042           | 0.0541     | 246,326 | 40               |
| Rewighted (pre-period age-group raking)                  | 2.597*         | 1.050 | 0.0178           | 0.0539     | 242,552 | 40               |
| Married women only                                       | 2.743          | 1.512 | 0.0775           | 0.1119     | 173,934 | 39               |
| Marital status $\times$ Post $\times$ WFH_Exposure added | 2.191          | 1.273 | 0.0932           | 0.1254     | 246,326 | 40               |
| <i>Sample</i>                                            |                |       |                  |            |         |                  |
| Unswapped occupations only                               | 4.309**        | 1.192 | 0.0011           | 0.0201     | 132,046 | 30               |
| Excluding survey year 2023                               | 2.979*         | 1.170 | 0.0151           | 0.0739     | 208,833 | 39               |
| Leave-one-occupation-out (40 refits): range              | [2.415, 3.580] |       | [0.0007, 0.0167] |            |         | 39               |
| <i>Occupational sorting</i>                              |                |       |                  |            |         |                  |
| Pre-period cell exposure as regressor, per SD            | 0.139          | 0.159 | 0.3823           |            | 247,931 | 245              |
| Exposure score as outcome: Mother $\times$ Post          | 0.001          | 0.003 | 0.6996           |            | 246,326 | individual       |
| Top-quartile indicator as outcome: Mother $\times$ Post  | 0.004          | 0.007 | 0.5433           |            | 246,326 | individual       |
| <i>Outcome coding</i>                                    |                |       |                  |            |         |                  |
| Irregular-hours codes dropped, not imputed               | 3.056**        | 1.059 | 0.0064           | 0.0573     | 232,785 | 40               |
| Full-time indicator ( $\geq 35$ hours)                   | 0.096**        | 0.029 | 0.0020           | 0.0466     | 246,326 | 40               |
| Long-hours indicator ( $\geq 40$ hours)                  | 0.107**        | 0.038 | 0.0075           | 0.0529     | 246,326 | 40               |
| <i>Inference on the primary estimate</i>                 |                |       |                  |            |         |                  |
| Two-way clustering (ind., occ.)                          | 3.224**        | 1.022 | 0.0031           |            | 246,326 | 40 $\times$ ind. |
| Wild cluster bootstrap ( $B = 9,999$ )                   | 3.224          | 1.022 | 0.0558           |            | 246,326 | 40               |
| Permutation test (999 reassignments)                     | 3.224          |       | 0.0320           |            | 246,326 | 40               |

Unless a row says otherwise: the triple interaction Mother  $\times$  Post  $\times$  WFH\_Exposure from equation (2), women working in the reference week, standard controls, calibrated exposure, clustered by occupation with the cluster count in the last column,  $p$  the analytic  $t(G - 1)$   $p$ -value and  $p$  (boot) the wild cluster bootstrap  $p$ -value of the same coefficient (Rademacher weights, null imposed, 9,999 draws), blank where the row is not a one-way occupation-clustered refit of a single model (the two-way-clustered, leave-one-out, sorting and cell-index rows). The men-only row calibrates the index on men’s realized 2022–23 WFH shares; the teaching-only row moves ISCO 23 alone from its external score to its realized share and leaves the other thirty-nine at the external score; the married-only row drops marital status from the controls; the marital-interacted row adds marital status  $\times$  Post  $\times$  WFH\_Exposure and its two-way terms; the swapped-occupation rows keep the external index as regressor on all forty occupations and add a Swapped indicator for the ten calibrated occupations with the full Mother  $\times$  Post structure, showing both triple interactions from that one model; the unswapped-occupations row drops those ten; the absentees row adds employed women absent from work in the reference week at their reported usual hours, on 2018–2023 only, since the 2017 file recorded absentees’ hours as zero. The leave-one-occupation-out row gives the range of the forty refits with one occupation removed (Appendix Figure A3) and the range of their  $p$ -values. Under *Occupational sorting*, the first row is the hours DDD with the pre-period demographic-cell index of Section 3.2 as regressor, reported per standard deviation of that index and clustered on the demographic cell; the next two are differences-in-differences with the occupation-level score, and an indicator for its top quartile, as the dependent variable, clustered by individual. The two indicator outcomes under *Outcome coding* are linear probability models on bin-crossing outcomes; the top-bin rows recode the open 60-plus bin, assigned 78.5 hours in the main coding, to 60 hours, and estimate a linear probability model for being in that bin. The wild-bootstrap row reports the analytic coefficient and standard error with the bootstrap  $p$ -value; the permutation row reports the permutation  $p$ -value, which has no standard error.

## 6.5 Who responded: child age, population group, and fathers

If remote work relaxes a time constraint that binds through child care, the response should be largest for mothers of young children and smallest for mothers of teenagers, and it is. Table 4 re-estimates the DiD and DDD for mothers whose youngest child is aged 0–4, 5–9, 10–14 or 15–17, each against the same childless women. The average effect is a null in every group, and the exposure gradient declines monotonically with the child’s age, from 3.656\*\* hours per unit of exposure (SE 1.079) for mothers of under-fives to 1.862 (SE 0.935) for mothers of 15–17-year-olds, half as large. The four regressions share a control group and the same forty clusters, so their intervals overlap and are not independent, and on the wild cluster bootstrap none of the four is significant at the 5% level (bootstrap  $p$  from 0.06 for the under-fives to 0.10 for the oldest group); the pattern is a gradient consistent with the mechanism rather than a set of statistically separated effects, and the estimate is not carried by mothers of teenagers.

Table 5 splits the sample by population group and re-estimates the models on men. Jewish women reproduce the pooled pattern (bootstrap  $p = 0.06$ ). Arab women’s gradient is nominally twice as large but statistically indistinguishable from it ( $z = 0.942$ ,  $p = 0.35$ ) on a subsample nine times smaller, and its bootstrap  $p$  is 0.14: imprecise rather than a documented heterogeneous effect.

For men, with “mother” read as “father,” the gradient has the opposite sign,  $-1.762$  (SE 1.015, analytic  $p = 0.09$ , bootstrap  $p = 0.15$ ), and is distinguishable from the mothers’ ( $z = 3.461$ ,  $p = 0.0005$ ). Fathers in teleworkable occupations are exposed to remote work exactly as mothers are (Section 5.5), so the comparison is not a placebo and a null was never the prediction. What it shows is that a macro trend common to all workers in teleworkable jobs cannot produce a positive gradient for mothers beside a negative one for fathers, and that opposite gradients are what within-household reallocation would look like if mothers’ additional paid hours were matched by fathers’ additional care hours. We report that as a reading the data are consistent with, not a result the design identifies: mothers and fathers are not linked in the data, and the fathers’ coefficient is itself imprecise.

## 6.6 The extensive margin

Appendix Table A3 reports equation (1) with employment as the outcome. The DiD is  $-0.0053$  (SE 0.0052), an informative null with a standard error of half a percentage point whose pre-period coefficients pass their joint test. The employment DDD with the cell-based index is a null of a different kind: the design could detect a triple interaction no smaller than 1.43 percentage points per standard deviation of the cell index, and against the US estimate of 0.78 percentage points per 10% rise in WFH (Harrington and Kahn, 2025) it is underpowered by a factor of two to three. The shortfall is structural, a binary

Table 4: Hours DiD and DDD by age of the youngest child

| Sample of mothers         | DiD: Mother $\times$ Post |       |         | DDD: Mother $\times$ Post $\times$ WFH_Exposure |       |            |         |
|---------------------------|---------------------------|-------|---------|-------------------------------------------------|-------|------------|---------|
|                           | Coef.                     | SE    | $N$     | Coef.                                           | SE    | $p$ (boot) | $N$     |
| All mothers (Table 2)     | 0.228                     | 0.181 | 251,857 | 3.224**                                         | 1.022 | 0.0558     | 246,326 |
| Youngest child aged 0–4   | 0.127                     | 0.211 | 158,424 | 3.656**                                         | 1.079 | 0.0551     | 154,499 |
| Youngest child aged 5–9   | 0.162                     | 0.253 | 130,782 | 3.211*                                          | 1.311 | 0.0944     | 127,419 |
| Youngest child aged 10–14 | 0.159                     | 0.278 | 123,280 | 2.741*                                          | 1.525 | 0.0935     | 120,084 |
| Youngest child aged 15–17 | 0.561                     | 0.348 | 107,061 | 1.862*                                          | 0.935 | 0.1020     | 104,079 |

Weekly hours, reference-week workers, standard controls; each row compares the named group of mothers with all childless women, so the rows are not independent samples. DiD clustered by individual; DDD clustered by occupation, calibrated exposure, with  $p$  (boot) the wild cluster bootstrap  $p$ -value of the triple interaction (9,999 Rademacher draws, null imposed). The first row repeats Table 2. No coefficient is dropped for collinearity in any column.

Table 5: Hours DiD and DDD by subgroup, and the fathers’ comparison

| Subgroup                     | DiD: Mother $\times$ Post |       |         | DDD: Mother $\times$ Post $\times$ WFH_Exposure |       |            |         |
|------------------------------|---------------------------|-------|---------|-------------------------------------------------|-------|------------|---------|
|                              | Coef.                     | SE    | $N$     | Coef.                                           | SE    | $p$ (boot) | $N$     |
| All women (primary)          | 0.228                     | 0.181 | 251,857 | 3.224**                                         | 1.022 | 0.0558     | 246,326 |
| Jewish women                 | 0.302                     | 0.203 | 214,449 | 2.847**                                         | 0.913 | 0.0643     | 209,727 |
| Arab women                   | −0.072                    | 0.479 | 24,975  | 5.507*                                          | 2.674 | 0.1438     | 24,307  |
| Men (fathers vs. childless)  | −0.400*                   | 0.188 | 261,543 | −1.762*                                         | 1.015 | 0.1494     | 251,779 |
| 95% CI, Jewish               | [−0.095, 0.699]           |       |         | [1.058, 4.635]                                  |       |            |         |
| 95% CI, Arab                 | [−1.010, 0.866]           |       |         | [0.266, 10.748]                                 |       |            |         |
| Jewish vs. Arab, $z$ ( $p$ ) | 0.720 (0.471)             |       |         | −0.942 (0.346)                                  |       |            |         |
| Women vs. men, $z$ ( $p$ )   | 2.406 (0.016)             |       |         | 3.461 (0.001)                                   |       |            |         |

Weekly hours, reference-week workers, standard controls. DiD clustered by individual; DDD clustered by occupation, with  $p$  (boot) the wild cluster bootstrap  $p$ -value of the triple interaction (9,999 Rademacher draws, null imposed). Men: “Mother” read as “Father,” a comparison population rather than a placebo (Section 5.5). Two-sample  $z$ -tests as described there. Confidence intervals use the normal critical value. The Arab-women DiD drops the religion category ‘other’, constant on that sample; the Arab-women DDD drops the religion category ‘other’, constant on that sample; no other column drops a coefficient.

outcome and a regressor with a standard deviation of 0.07, and re-clustering, combining the two margins and refining the cell partition did not close it. The employment DDD is therefore *uninformative*, evidence neither for nor against a WFH-mediated employment effect, and the two nulls should not be read as one verdict.

## 7 Discussion

Among mothers who remain in work there is no post-2021 hours effect on average, and a large one in teleworkable occupations: a triple interaction of about 3.2 hours per unit of exposure and an implied change at zero exposure that is if anything negative. This is consistent with WFH access relaxing a binding time constraint for mothers in teleworkable jobs, the flexibility mechanism (Goldin, 2014), and hard to reconcile with a general recovery common to all women, which would have raised hours across occupations rather than only where the work can be done from home. The supporting evidence is a saturated specification that changes nothing, a top-quartile contrast significant on its own, a leave-one-out range that never leaves the headline’s standard error, a monotone child-age gradient, an opposite-signed fathers’ gradient and a selection correction that barely moves the bounds. Those checks are not independent confirmations: every one of them is estimated on the same forty occupation clusters as the headline, and where a wild cluster bootstrap  $p$ -value is available it lies above the 5% level, from 0.06 for the youngest child-age group to 0.15 for the fathers’ gradient, with the headline itself at 0.056 (Tables 3–5). What the checks establish is that the gradient is not an artifact of one specification, one occupation, one sample or one outcome coding; how precisely it is estimated remains the question the three inference procedures answer differently. The gradient is identified off a concrete contrast: teaching and clerical support, which Israeli practice kept in person, placed low against the professional occupations placed high. Among the latter, science and engineering professionals (ISCO 21) carry the most weight, since removing them gives the leave-one-out minimum of 2.415, while ICT professionals, high on every exposure measure, work slightly against the gradient, since removing them raises it to 3.287 (Appendix Figure A3).

The magnitude is economically meaningful where it applies. For a mother at the top quartile’s mean exposure the regression implies a post-2021 change of about 1.4 hours a week ( $-0.54 + 3.22 \times 0.59$ ), and the raw top-quartile difference-in-differences of Figure 1 is 1.55. That 1.4 is the fitted change *at* top-quartile exposure, distinct from the 1.71-hour fitted contrast *between* the quartile means. The penalty it is set against must be the penalty in those same jobs, not the average one: column (2) of Table 2 puts the pre-period motherhood gap at  $-1.45 - 1.30 \times 0.59 \approx -2.2$  hours at that exposure, so mothers in the most teleworkable jobs recovered roughly three fifths of their penalty after 2021. At the bottom quartile’s mean exposure of 0.06 the implied change is  $-0.35$  hours:

mothers in the least teleworkable jobs recovered none of it. Commuting time alone does not account for the size. Workers save about 72 minutes per day worked from home and return roughly 40% of that time to their jobs, the rest going to leisure and care (Aksoy et al., 2023), so two remote days a week add about one hour of paid work, not two and a half. And the 1.4 hours is an average over *all* mothers in top-quartile jobs, and even in those occupations only a minority of workers are at home in the reference week (the pooled shares by occupation in Appendix Table A1), so the response per mother who actually works from home is several times larger than the commute she saves. That points to the channel of the framework in Section 2: flexibility in *when* hours are worked, hours split around care, rather than travel time returned to work. The timing is consistent with a multi-year adjustment: the DDD event study rises from 3.0 in 2021 to 4.0 in 2023 (on the 2023 year, see Section 8).

A general recovery is not the only alternative to that mechanism; three demand-side accounts also fit a gradient by teleworkability. Employers in professional and ICT occupations may have changed the contracts they offer mothers after 2021, with no change in mothers' constraints; the saturated specification absorbs any occupation-by-year shock common to mothers and childless women, so this needs a change aimed at mothers in teleworkable occupations specifically. The 2021–22 technology hiring boom raised hours in exposed occupations for everyone; the fathers' opposite-signed gradient rules out a shock common to all workers in those occupations, but not one that fell on mothers alone. And usual hours are self-reported: if mothers working from home report the hours they are available while fathers report the hours they work, the two gradients would diverge as observed with no change in labor supplied. The child-age gradient does not separate the two readings: teleworking mothers of young children report more interruptions to the workday (Pablonia and Vernon, 2022), so the gap between hours available and hours worked would itself be largest for mothers of infants, the pattern the gradient shows. The fathers' gradient and the saturated specification are the arguments that remain. None of the three is excluded by the design, and we read the exposure gradient as the time-constraint mechanism's signature rather than its proof.

Why hours and not employment? The employment DDD is not a failed test of the mechanism but an uninformative one (Section 6.6). Substantively the two results are coherent: the effect operates on *how much* mothers in teleworkable jobs work, not on *whether* they are employed, which is a leading channel of the child penalty (Kleven, Landais and Søgaard, 2019), and it is where our findings depart from the US evidence on employment and income (Harrington and Kahn, 2025). The policy reading is correspondingly narrow: WFH availability matters for how intensively mothers who already hold *teleworkable* jobs can work, not for whether mothers in general hold jobs; and access to those jobs is itself unequally distributed across mothers, so the population-wide gain is smaller than the gradient alone suggests and cannot be sized without representative weighting.

## 8 Limitations

**Identification and exposure.** The triple difference’s own pre-trend test (Section 6.3) shows no visible violation but lacks the power to rule one out. The exposure measure that delivers the headline draws on post-period Israeli data for its calibration. The result requires reclassifying teaching, which the external index places in the treated tail and Israeli schools kept in person; the fully pre-treatment external index yields a positive and insignificant estimate, and the checks of Section 6.3 bound how much of the gradient the reclassification carries. The sorting check finds no post-2021 movement of mothers across exposure levels but bounds the effect only imprecisely on a regressor that a woman’s post-2021 occupation cannot move. Exposure is assigned at the two-digit occupation level, forty groups, whereas sorting across firms, even within the same industry, is a first-order driver of Israeli gender gaps (Buzaglo-Baris, 2023). Assigning every woman her occupation’s mean teleworkability is Berkson-type error, which does not by itself bias the slope; error in the occupation score itself attenuates the triple interaction toward zero; and if mothers sort into more or less teleworkable jobs *within* an occupation, the bias can run in either direction. The calibrated score also mixes two scales (Appendix B). Occupation codes masked by the CBS for disclosure control are dropped when building exposure; a proxy check finds a small, statistically detectable difference in remote work between masked and unmasked rows (Appendix B). How precisely the headline is estimated is the range of the three inference procedures in Section 6.2, not the analytic  $p$ -value, and the same holds for the supporting rows of Tables 3–5, whose stars come from the analytic test and whose bootstrap  $p$ -values mostly lie above 0.05. The permutation test itself assumes that the forty exposure scores are exchangeable across occupations under its null.

**The narrowed hours estimand.** Hours are usual hours among women who were working, not among all women holding a job, and roughly 10% of each year’s employed sample is excluded on that basis. The exclusion is not innocuous for the triple interaction because absence is related to exposure: it is highest in the least teleworkable quartile, and among mothers it fell after 2021 in the top quartile while rising in the bottom one (Appendix A). The bounds of Section 5.4 therefore define selection on the observed-hours sample; they handle only *excess* selection under monotonicity and found little to correct, which is not the same as the concern being absent.

**Sample and period.** The 2023 coefficient is the largest in both event studies, and the 2023 survey year includes the quarter in which the war that began on 7 October 2023 disrupted the labor market: reserve call-ups, which bear on the fathers’ comparison, and school closures and evacuations, which bear on mothers’ hours. The estimate survives dropping the year and 2021 and 2022 are individually significant on the triple difference,

but the 2023 point should be held more loosely. Early 2021 likewise included a third lockdown, so the post-period is “post-acute” more cleanly in its later years than in its first. The control group is any woman aged 25–59 with no child under 17 in the household, so it includes women whose children are grown and women who become mothers within the window, as in the US study (Harrington and Kahn, 2025); if part of it responds to the treatment as mothers do, the DiD is biased toward zero. Age enters as a categorical code because the extract has no continuous age, and mothers and non-mothers are imbalanced on it in a way that tracks exposure (Appendix Table A4); the age-interacted and reweighted specifications of Section 6.4 address this and the estimate survives both. Marital status is imbalanced in the same direction, and the two marital checks of that section leave the gradient positive but only marginally significant.

**Survey weights.** CBS survey weights are intentionally not applied in any outcome regression; reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index.

## 9 Conclusion

Using Israeli survey data, we ask whether the expansion of remote work coincided with a change in the motherhood penalty in usual weekly hours. The average change among women observed working is small, whereas the post-2021 change in the mother/comparison gap increases with an occupation’s calibrated WFH exposure. The strongest contrast is at the top of the exposure distribution and among mothers with younger children. Those patterns are consistent with greater scheduling flexibility helping some already employed mothers supply more paid hours.

The result is conditional and does not establish a universal benefit from remote work. The exposure score partly uses post-period WFH information; the wholly external score yields a weaker result, pre-trend tests have limited power, and inference varies with the procedure. Employment is an important separate margin, but its exposure interaction is too imprecise here to settle the question. The data also contain no earnings amount. Stronger evidence would link hours with earnings and household time use, measure exposure before treatment in the Israeli setting, and estimate women’s and men’s responses together on a design that better separates access to WFH from occupation choice.

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## A Data and sample details

**Hours coding.** The CBS records usual weekly hours in bins; each bin is assigned its median (4, 11, 18, 25.5, 32, 37, 42, 47, 54.5 and 78.5 hours) and the two irregular-hours codes are imputed from observed medians within the same period. Mid-distribution bins are about five hours wide, so an effect smaller than one bin width, such as the headline estimate, is best read as a shift in the share of women crossing a bin boundary. In the 2017 file, employed respondents absent from work in the reference week had their usual hours recorded as zero, whereas from 2018 onward the same respondents were assigned a real value; left uncorrected, this enters the pre-period as roughly 5,200 spurious zeros, and because absence is more common among mothers (12.4% of employed mothers against 7.3% of childless women) they fall unevenly across the comparison the paper makes. Defining hours on the reference-week workers in *every* year costs about 10% of each year’s employed sample, makes the hours population identical across years, and narrows the estimand to usual hours among women who were working.

**Reference-week absence and exposure.** Absence is related to exposure. Among employed women with a matched occupation, by the quartiles of Figure 1, the absence shares are 14.4%, 8.5%, 9.5% and 9.6%. Among mothers the top-quartile share fell from 12.1% to 10.4% after 2021 and the bottom-quartile share rose from 15.5% to 16.9%; among childless women the top-quartile share is 6.1% in both periods. Movement across the absence margin is therefore part of what selection into the observed-hours sample means, which is why the bounds of Appendix C define selection on that sample.

**Panel structure.** The survey is a rotating panel. The 258,175 reference-week workers of Table 1 come from 64,602 distinct respondents, about four observations each and two to three within a single survey year, with 96% of rows belonging to someone who appears more than once; the regressions use 251,857 of those rows from 63,199 respondents, the difference being the 6,318 rows whose education code is missing (education is the only control with missing values). Treating the observations as independent would understate standard errors by a factor of roughly 1.8, which is why every DiD and every descriptive

interval is clustered by individual. The panel does not support within-person identification: of the 80,560 women in the full sample, 1.9% are interviewed on both sides of the pre/post boundary, so an individual fixed-effects design was considered and rejected.

**Hours year by year.** Panel (a) of Figure A1 unpools the pre-period. The raw mother-minus-non-mother gap is  $-1.37$  hours in 2017,  $-1.48$  in 2018 and  $-1.67$  in 2019, a widening of 0.3 hours over two years that is monotone but within sampling error, each year lying inside the others' confidence bounds. The post-period path is  $-1.37$ ,  $-1.53$  and  $-0.80$ , not monotone and not individually distinguishable, so the timing reading in the Discussion rests on the event studies of Figure 2, not on this series.

**Commuting.** Mothers were the *more* mobile group in every year, working outside their locality of residence 48.2% of the time in 2019 against 46.1% of childless women, and their commuting fell less in 2021 (1.5 against 3.9 percentage points, a difference of 2.39 points, SE 1.17). This is consistent with remote work relaxing a time constraint for women who were already commuting; it is descriptive evidence only, and no specification uses locality of work as an outcome.

## B Exposure construction

**The external score.** The Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) reaches the two-digit ISCO-08 level through the BLS crosswalk between ISCO-08 and the 2010 SOC (Bureau of Labor Statistics, 2012). Each O\*NET-SOC occupation in their published classification is matched to every ISCO-08 unit group the crosswalk lists for its six-digit SOC code, and the score of a sub-major group is the unweighted mean of their binary indicator over the matched pairs; armed-forces groups have no counterpart in their data and carry no score. The replication code re-derives the forty scores (Table A1) from the two public source files and checks them against the file the pipeline reads. Teaching (ISCO 23) is the clearest case of a US task-based score misclassifying Israeli practice: it scores 0.966 on the external index, yet Israeli schools, closed and reopened in stages through 2020 and early 2021 (Somekh et al., 2021; Hale et al., 2021), were fully in person throughout the 2022–23 survey years, and teachers did not work from home.

**The calibration rule.** The external score is replaced by the realized 2022–23 Israeli WFH share only where the gap between the two is large (greater than 0.5) *and* distinguishable from sampling noise by a one-sided test at the 95% level with a standard error clustered by individual. Ten of the forty two-digit occupations are swapped: teaching falls from 0.966 to 0.069, clerical support (ISCO 41) from 1.000 to 0.098, ICT professionals (ISCO 25) from 1.000 to 0.360. The realized shares are computed on men and women

together; Table 3 re-derives them from men alone. The realized index used as a bound is the realized Israeli WFH share by occupation, anchored to 2021 (no 2020 extract exists) with a minimum of 200 observations per occupation.

**Masked occupation codes.** The CBS masks some two-digit occupation codes for disclosure control, and those rows carry no exposure score and drop out of the hours DDD. A partially masked code still identifies the one-digit major group, so the check compares realized WFH between masked and unmasked employed women within the same major group in 2022–23. Among professionals (major group 2), the masked rows work from home at a share of 0.29 against 0.16 for the unmasked ones, on 207 masked observations; in the other major groups the masked rows are fewer and the gaps are small or negative.

**Fit.** Across the forty occupations, on men and women pooled over 2021–23, the calibrated score has a correlation of 0.53 with the share who worked from home in the reference week and an occupation-size-weighted slope of 0.40 (SE 0.09), so it is relevant. The raw external index fits better in levels (correlation 0.74), because swapping ten occupations onto realized shares while leaving thirty on task-based probabilities mixes two scales. The calibration’s contribution is not a better cross-sectional fit but the removal of ten grossly mis-scored occupations from the treated tail. Table A1 lists, for each occupation, the external score, the realized 2022–23 share the calibration tests it against, whether the score was swapped, the calibrated score the hours DDD uses, and the realized reference-week share over 2021–23; panel (b) of Figure A1 plots the last against the calibrated score.

**Sensitivity to the swap threshold.** The calibration rule swaps an occupation when its external-minus-realized gap exceeds 0.5 by more than sampling noise. Table A2 refits the hours DDD with that threshold moved from 0.30 to 0.90. Lowering it swaps more occupations and the estimate grows noisier; from 0.75 upward only teaching and clerical support are swapped, and at 0.90, above the largest gap, nothing is swapped and the row is the external-index estimate of Table 3. The estimate is positive at every threshold, and the step from the external index to any positive threshold is the reclassification of teaching, which Table 3’s teaching-only row isolates.

## C The selection correction

Let  $s_{ab} = \Pr(\text{hours observed} \mid \text{Mother} = a, \text{Post} = b)$ , the probability of being employed and working in the reference week. Under parallel trends in selection the counterfactual for post-period mothers is  $s_{11}^* = s_{10} + s_{01} - s_{00}$ . If  $s_{11}$  exceeds it, the excess share  $1 - s_{11}^*/s_{11}$  is trimmed from the (Mother = 1, Post = 1) hours distribution, from the top for the lower bound and from the bottom for the upper bound, and the model is re-estimated on each

Table A1: External, realized and calibrated WFH exposure by two-digit occupation

| ISCO | Occupation (ISCO-08 sub-major group)                           | Ext.  | Real. 22–23 | Swap | Calib. | Ref.-wk 21–23 | <i>N</i> |
|------|----------------------------------------------------------------|-------|-------------|------|--------|---------------|----------|
| 11   | Chief executives and legislators                               | 0.900 | 0.155       | yes  | 0.155  | 0.320         | 2,855    |
| 12   | Administrative and commercial managers                         | 0.892 | 0.143       | yes  | 0.143  | 0.336         | 3,859    |
| 13   | Production and specialised services managers                   | 0.638 | 0.097       | yes  | 0.097  | 0.199         | 9,990    |
| 14   | Hospitality, retail and other services managers                | 0.773 | 0.066       | yes  | 0.066  | 0.132         | 2,827    |
| 21   | Science and engineering professionals                          | 0.644 | 0.210       |      | 0.644  | 0.349         | 13,228   |
| 22   | Health professionals                                           | 0.101 | 0.059       |      | 0.101  | 0.088         | 9,665    |
| 23   | Teaching professionals                                         | 0.966 | 0.069       | yes  | 0.069  | 0.184         | 20,496   |
| 24   | Business and administration professionals                      | 0.943 | 0.233       | yes  | 0.233  | 0.449         | 10,479   |
| 25   | ICT professionals                                              | 1.000 | 0.360       | yes  | 0.360  | 0.615         | 14,638   |
| 26   | Legal, social and cultural professionals                       | 0.678 | 0.215       |      | 0.678  | 0.371         | 13,344   |
| 31   | Science and engineering associate professionals                | 0.144 | 0.059       |      | 0.144  | 0.106         | 3,926    |
| 32   | Health associate professionals                                 | 0.057 | 0.080       |      | 0.057  | 0.084         | 2,203    |
| 33   | Business and administration associate professionals            | 0.684 | 0.123       | yes  | 0.123  | 0.223         | 22,415   |
| 34   | Legal, social and cultural associate professionals             | 0.577 | 0.142       |      | 0.577  | 0.206         | 5,480    |
| 35   | ICT technicians                                                | 0.750 | 0.244       |      | 0.750  | 0.380         | 2,491    |
| 41   | General and keyboard clerks                                    | 1.000 | 0.098       | yes  | 0.098  | 0.144         | 4,846    |
| 42   | Customer services clerks                                       | 0.292 | 0.096       |      | 0.292  | 0.159         | 3,875    |
| 43   | Numerical and material recording clerks                        | 0.526 | 0.071       |      | 0.526  | 0.152         | 4,948    |
| 44   | Other clerical support workers                                 | 0.625 | 0.083       | yes  | 0.083  | 0.161         | 1,049    |
| 51   | Personal service workers                                       | 0.271 | 0.146       |      | 0.271  | 0.144         | 7,808    |
| 52   | Sales workers                                                  | 0.200 | 0.050       |      | 0.200  | 0.063         | 8,704    |
| 53   | Personal care workers                                          | 0.300 | 0.046       |      | 0.300  | 0.045         | 17,206   |
| 54   | Protective services workers                                    | 0.100 | 0.008       |      | 0.100  | 0.011         | 3,969    |
| 61   | Market-oriented skilled agricultural workers                   | 0.182 | 0.027       |      | 0.182  | 0.051         | 1,325    |
| 62   | Market-oriented skilled forestry, fishery and hunting workers  | 0.179 | 0.000       |      | 0.179  | 0.045         | 22       |
| 63   | Subsistence farmers, fishers, hunters and gatherers            | 0.000 | 0.375       |      | 0.000  | 0.375         | 8        |
| 71   | Building and related trades workers                            | 0.027 | 0.008       |      | 0.027  | 0.016         | 5,208    |
| 72   | Metal, machinery and related trades workers                    | 0.000 | 0.008       |      | 0.000  | 0.013         | 3,982    |
| 73   | Handicraft and printing workers                                | 0.138 | 0.118       |      | 0.138  | 0.129         | 666      |
| 74   | Electrical and electronic trades workers                       | 0.000 | 0.022       |      | 0.000  | 0.042         | 3,424    |
| 75   | Food, wood, garment and other craft workers                    | 0.073 | 0.122       |      | 0.073  | 0.109         | 2,304    |
| 81   | Stationary plant and machine operators                         | 0.000 | 0.019       |      | 0.000  | 0.017         | 2,407    |
| 82   | Assemblers                                                     | 0.000 | 0.012       |      | 0.000  | 0.015         | 1,166    |
| 83   | Drivers and mobile plant operators                             | 0.237 | 0.006       |      | 0.237  | 0.006         | 8,480    |
| 91   | Cleaners and helpers                                           | 0.000 | 0.004       |      | 0.000  | 0.003         | 3,474    |
| 92   | Agricultural, forestry and fishery labourers                   | 0.000 | 0.007       |      | 0.000  | 0.016         | 192      |
| 93   | Labourers in mining, construction, manufacturing and transport | 0.120 | 0.006       |      | 0.120  | 0.008         | 2,659    |
| 94   | Food preparation assistants                                    | 0.000 | 0.013       |      | 0.000  | 0.011         | 649      |
| 95   | Street and related sales and service workers                   | 0.000 | 0.000       |      | 0.000  | 0.043         | 23       |
| 96   | Refuse workers and other elementary workers                    | 0.250 | 0.022       |      | 0.250  | 0.024         | 493      |

Ext.: the Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) at the two-digit ISCO-08 level, via the crosswalk described above. Real. 22–23: share of employed respondents (men and women, all ages in the extract) who usually work from home in 2022–23, the calibration’s benchmark; Swap: the calibration replaced the external score with it. Calib.: the score the hours DDD uses. Ref.-wk 21–23: share who worked from home in the survey reference week over 2021–23, men and women pooled, with *N* the pooled employed sample carrying that item. Codes 1–3 (armed forces) carry no external score and are not in the sample.

Table A2: Hours DDD by calibration gap threshold

| Gap threshold | Occupations swapped | Coefficient | SE    | $p$      | Clusters |
|---------------|---------------------|-------------|-------|----------|----------|
| 0.30          | 15                  | 3.930       | 2.819 | 0.1712   | 40       |
| 0.35          | 15                  | 3.930       | 2.819 | 0.1712   | 40       |
| 0.40          | 15                  | 3.930       | 2.819 | 0.1712   | 40       |
| 0.45          | 12                  | 4.239***    | 1.177 | 0.0009   | 40       |
| 0.50          | 10                  | 3.224**     | 1.022 | 0.0031   | 40       |
| 0.55          | 8                   | 3.096**     | 0.939 | 0.0021   | 40       |
| 0.60          | 7                   | 2.902***    | 0.729 | 0.0003   | 40       |
| 0.65          | 6                   | 2.475***    | 0.673 | 0.0007   | 40       |
| 0.70          | 4                   | 2.657***    | 0.565 | < 0.0001 | 40       |
| 0.75          | 2                   | 2.319***    | 0.504 | < 0.0001 | 40       |
| 0.80          | 2                   | 2.319***    | 0.504 | < 0.0001 | 40       |
| 0.85          | 2                   | 2.319***    | 0.504 | < 0.0001 | 40       |
| 0.90          | 0                   | 0.672       | 0.884 | 0.4517   | 40       |

Mother  $\times$  Post  $\times$  WFH\_Exposure from equation (2), women working in the reference week, standard controls, clustered by occupation, with the exposure index recalibrated at the gap threshold in the first column (the significance test on the gap is unchanged). The 0.50 row is the headline of Table 2; a row with no occupation swapped is the external Dingel–Neiman index. Stars follow the paper’s legend on the analytic  $t(G - 1)$  test.

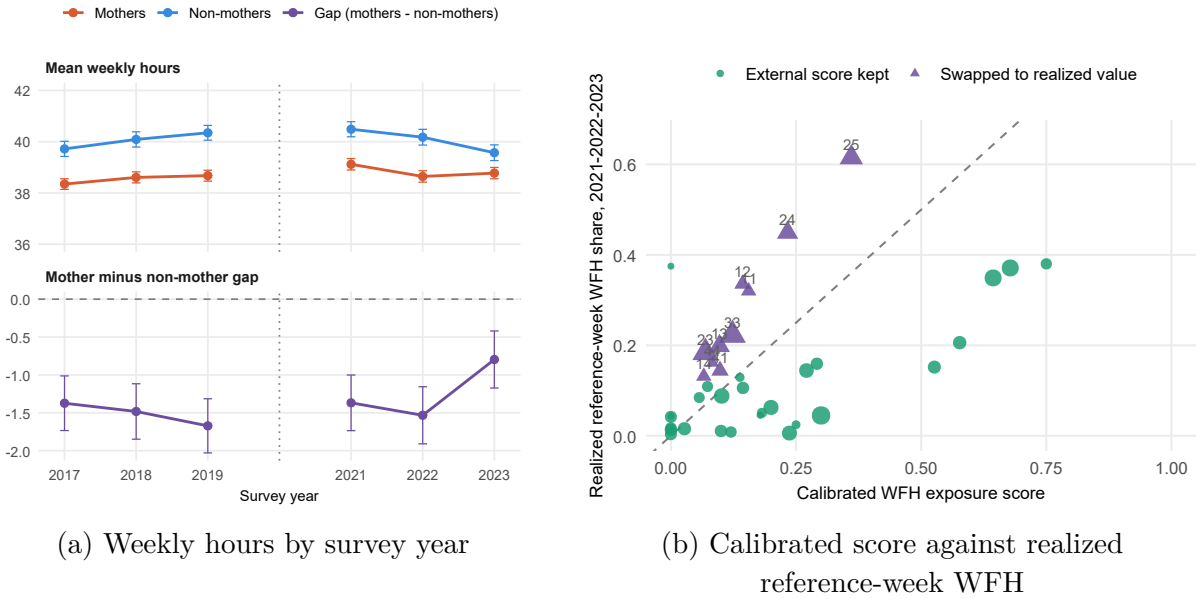


Figure A1: Weekly hours by survey year, and the exposure first stage  
*Notes:* Panel (a): women aged 25–59 who worked in the reference week, unweighted means with no controls; the upper panel plots each group’s mean weekly hours on an axis spanning 36 to 42 hours, the lower panel their difference, with 95% confidence intervals clustered by individual; the dotted rule marks the excluded 2020 survey year, and the series break there. Panel (b): one point per two-digit occupation, sized by the pooled employed sample in 2021–23; triangles are the ten occupations the calibration swapped to their realized value, labelled by ISCO-08 code; the dashed line is the 45-degree line. The swapped occupations sit above it because their calibrated value is the usual-place share, which is lower than the reference-week share everywhere; the thirty unswapped ones lie below it, the visual form of the scale-mixing described above.

trimmed sample. Trimming one tail assumes monotone selection, and the construction handles only excess selection in that cell. For the DDD the counterfactual and trim are computed within each quartile of the cell-based exposure index, which assumes selection depends on exposure only through that quartile, and equation (2) is refit on the recombined sample with the occupation-level regressor and occupation clustering. Because the parameter is partially identified, the Imbens–Manski interval for the identified set is reported with the bounds (Imbens and Manski, 2004).

For the DiD,  $s_{11} = 0.696$  against  $s_{11}^* = 0.683$ , and trimming the 1.9% excess gives bounds of  $-0.517$  (SE 0.178) and  $0.802$  (SE 0.179) around the 0.228 point estimate, with Imbens–Manski 95% interval  $[-0.810, 1.096]$ . For the DDD the excess is present in all four quartiles but modest, a trim of 0.80% to 4.48%, 1,396 of 72,820 post-period working mothers; the bounds are 3.126 (SE 1.144) and 3.443 (SE 1.022) around an untrimmed 3.232 (SE 1.054), which differs from Table 2’s 3.224 because this sample additionally requires a matched cell-based exposure, and the Imbens–Manski interval is  $[1.021, 5.324]$ .

## D Additional results

Figure A2 plots the reference distribution of the permutation test: under the sharp null the central 95% of cluster-robust  $t$ -statistics lie between  $-2.96$  and  $2.60$ , and the observed 3.15 is exceeded in absolute value by 31 of 999 draws. Table A3 reports the extensive-margin DiD and DDD discussed in Section 6.6. Table A4 reports the pre-period covariate balance between mothers and childless women within each exposure quartile of the hours DDD’s estimation sample, and Figure A3 the forty leave-one-occupation-out refits of the headline estimate.

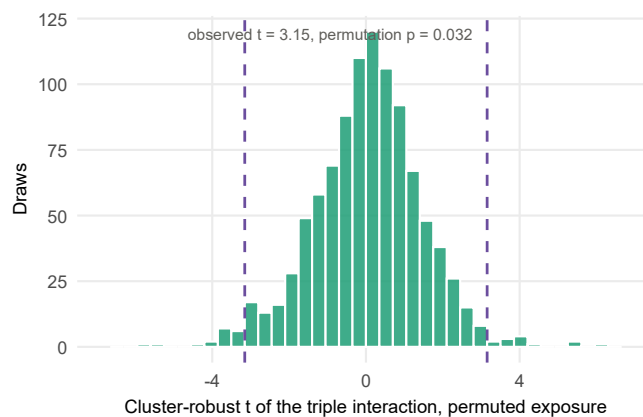


Figure A2: Permutation distribution of the hours DDD triple interaction

*Notes:* Histogram of the cluster-robust  $t$ -statistic of  $\text{Mother} \times \text{Post} \times \text{WFH\_Exposure}$  over 999 random reassignments of the forty exposure scores across occupations, with the same sample, controls and clustering as column (2) of Table 2. Dashed lines mark  $\pm$  the observed statistic (3.15);  $p = 0.032$  counts the observed draw (Phipson and Smyth, 2010).

Table A3: Extensive margin: employment DiD and DDD

|                                               | (1) DiD               | (2) DDD, cell-based exposure       |
|-----------------------------------------------|-----------------------|------------------------------------|
| Mother $\times$ Post $\times$ WFH_Exposure    |                       | −0.0257<br>(0.0725)                |
| Mother $\times$ Post                          | −0.0053<br>(0.0052)   | 0.0065<br>(0.0168)                 |
| Mother $\times$ WFH_Exposure                  |                       | −0.1404<br>(0.0803)                |
| Post $\times$ WFH_Exposure                    |                       | −0.1555**<br>(0.0566)              |
| WFH_Exposure                                  |                       | 0.0131<br>(0.0565)                 |
| Mother                                        | −0.0034<br>(0.0041)   |                                    |
| Post                                          | 0.0153***<br>(0.0043) |                                    |
| Constant                                      | 0.4926***<br>(0.0085) |                                    |
| MDE ( $\alpha = 0.05$ , power 0.80), per unit |                       | 0.2031                             |
| MDE per SD of exposure                        |                       | 0.0143                             |
| as % of baseline employment rate              |                       | 1.85%                              |
| Observations                                  | 364,784               | 355,984                            |
| $R^2$                                         | 0.227                 | 0.221                              |
| Clustering                                    | individual            | demographic cell ( $\approx 210$ ) |

Dependent variable: employment indicator; full sample of employed and non-employed women; standard controls in both columns, four decimals because the outcome is a probability. Column (2) adds Mother  $\times$  age-group interactions, so its Mother and Post main effects are not separately reported, and uses the demographic-cell index of Section 3.2, whose standard deviation on this sample is 0.0706; the per-SD MDE row is the interpretable one. Baseline employment rate 0.7737. No coefficient is dropped for collinearity in any column.

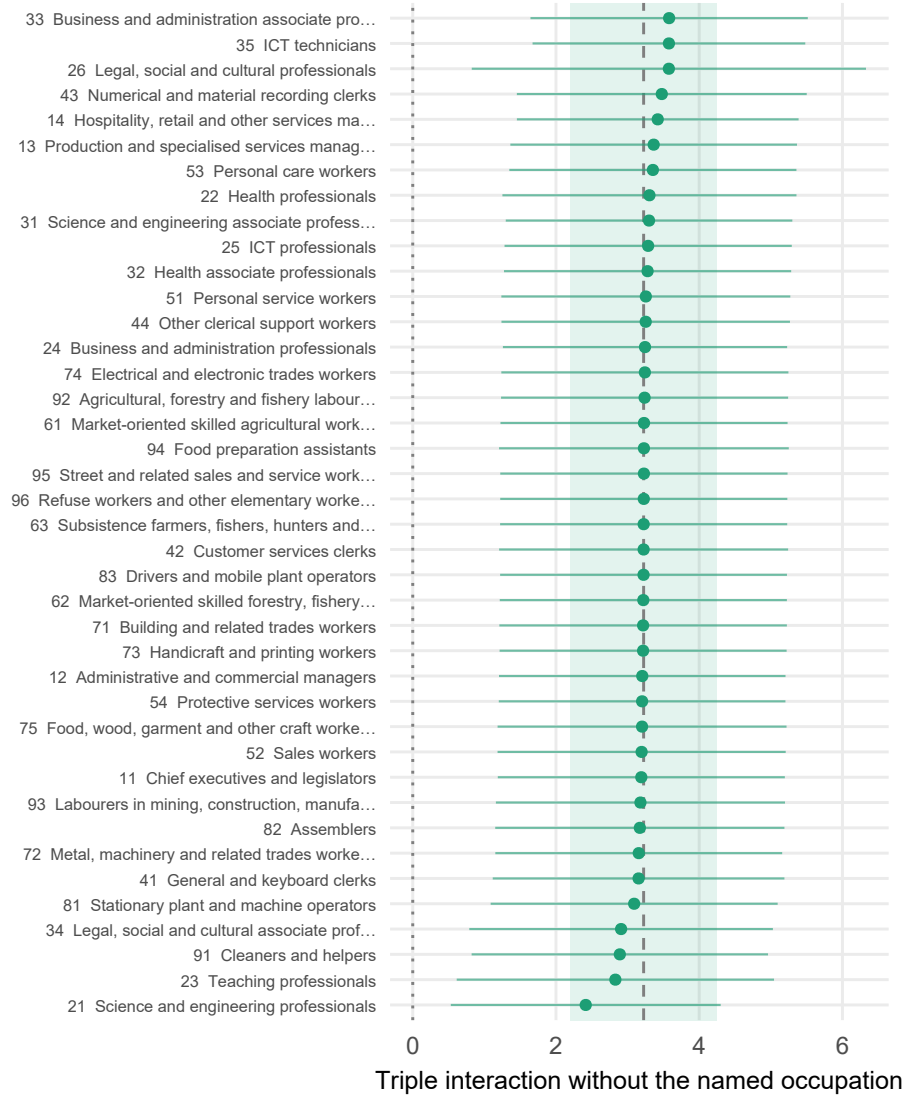


Figure A3: Leave-one-occupation-out: the hours DDD triple interaction with each occupation removed

*Notes:* Each point is  $\text{Mother} \times \text{Post} \times \text{WFH\_Exposure}$  from equation (2) re-estimated without the occupation named on the axis (column (2) of Table 2 on 39 occupations), with its 95% confidence interval clustered by occupation; rows are sorted by the estimate. The dashed line is the headline estimate on all forty occupations and the shaded band one headline standard error either side of it; the dotted line marks zero. Triangles would mark refits outside the band; there are none.

Table A4: Pre-period covariate balance by quartile of occupation-level WFH exposure

| Mothers minus childless women, pre-period | Q1                 | Q2                 | Q3                 | Q4               |
|-------------------------------------------|--------------------|--------------------|--------------------|------------------|
| Age-group code (CBS 3–7), mean            | −0.54***<br>(0.03) | −0.44***<br>(0.03) | −0.33***<br>(0.03) | 0.08*<br>(0.04)  |
| Married (%)                               | 32.2***<br>(1.1)   | 33.7***<br>(1.1)   | 41.5***<br>(0.9)   | 43.1***<br>(1.3) |
| Academic degree (%)                       | 10.8***<br>(1.2)   | 4.5***<br>(1.2)    | 1.5<br>(1.0)       | 1.8<br>(1.2)     |
| Below high school (%)                     | −3.2***<br>(0.6)   | −1.7***<br>(0.4)   | −3.3***<br>(0.5)   | −0.2<br>(0.2)    |
| Jewish household head (%)                 | 1.6<br>(1.0)       | 0.5<br>(0.7)       | 14.3***<br>(0.9)   | 3.1***<br>(0.8)  |
| Center or Tel Aviv district (%)           | −4.4***<br>(1.2)   | −3.4**<br>(1.2)    | −8.8***<br>(1.1)   | −4.5**<br>(1.4)  |
| Arab (%)                                  | 1.9*<br>(0.9)      | 0.2<br>(0.6)       | 3.7***<br>(0.6)    | −0.9<br>(0.6)    |
| Mothers                                   | 22,976             | 24,664             | 22,562             | 16,377           |
| Childless women                           | 11,420             | 11,586             | 17,244             | 8,352            |
| Occupations                               | 15                 | 7                  | 11                 | 6                |
| Exposure range                            | 0.00–0.10          | 0.10–0.14          | 0.14–0.30          | 0.36–0.75        |

Women working in the reference week in 2017–2019 with a matched two-digit occupation, the hours DDD’s pre-period estimation sample, split into the exposure quartiles of Figure 1. Each cell is the mother-minus-childless difference in the row variable within the quartile, with a standard error clustered by individual in parentheses; shares are in percentage points, the age-group code in code units (CBS codes 3–7 span ages 25–59 in five-year bands). Stars follow the paper’s legend on a normal reference. Unweighted.